

Homework 5

Problem 1

Use differentials to show that, for large n and small a , $\sqrt{n+a} - \sqrt{n} \cong \frac{a}{2\sqrt{n}}$.

Find the approximate value of $\sqrt{10^{26} + 5} - \sqrt{10^{26}}$.

Problem 2

(1)

Given $z = (x + y)^5$, $y = \sin 10x$, find dz/dx

(2)

Given $c = \sin(a - b)$, $b = ae^{2a}$, find dc/da .

Problem 3

(1)

If $z = xe^{-y}$ and $x = \cosh t$, $y = \cos s$, find $\partial z/\partial s$ and $\partial z/\partial t$.

(2)

If $w = e^{-r^2-s^2}$, $r = uv$, $s = u + 2v$, find $\partial w/\partial u$ and $\partial w/\partial v$.

(3)

If $w = x + y$ with $x^3 + xy + y^3 = s$ and $x^2y + xy^2 = t$, find $\partial w/\partial s$, $\partial w/\partial t$.

Problem 4

Find the point on $2x + 3y + z - 11 = 0$ for which $4x^2 + y^2 + z^2$ is a minimum.

Problem 5

Find the shortest distance from the origin to the line of intersection of the planes $2x + y - z = 1$ and $x - y + z = 2$.

Problem 6

In the partial differential equation

$$\frac{\partial^2 z}{\partial x^2} - 5 \frac{\partial^2 z}{\partial x \partial y} + 6 \frac{\partial^2 z}{\partial y^2} = 0$$

put $s = y + 2x$, $t = y + 3x$ and show that the equation becomes $\partial^2 z / \partial s \partial t = 0$. Following the method of solving (11.6), solve the equation.

$$(11.6) \quad \frac{\partial^2 F}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 F}{\partial t^2} = 4 \frac{\partial^2 F}{\partial r \partial s} = 0.$$

We can easily solve (11.6). We have

$$\frac{\partial^2 F}{\partial r \partial s} = \frac{\partial}{\partial r} \left(\frac{\partial F}{\partial s} \right) = 0,$$

that is, the r derivative of $\partial F / \partial s$ is zero. Then $\partial F / \partial s$ must be independent of r , so $\partial F / \partial s =$ some function of s alone. We integrate with respect to s to find $F = f(s) + \text{“const.”}$; the “constant” is a constant as far as s is concerned, but it may be any function of r , say $g(r)$, since $(\partial / \partial s)g(r) = 0$. Thus we find that the solution of (11.6) is

$$(11.7) \quad F = f(s) + g(r).$$

Problem 7

Suppose that $w = f(x, y)$ satisfies

$$\frac{\partial^2 w}{\partial x^2} - \frac{\partial^2 w}{\partial y^2} = 1.$$

Put $x = u + v$, $y = u - v$, and show that w satisfies $\partial^2 w / \partial u \partial v = 1$. Hence solve the equation.

Problem 8

For each of the following differential equations, separate variables and find a solution containing one arbitrary constant. Then find the value of the constant to give a particular solution satisfying the given boundary condition. Computer plot a slope field and some of the solution curves.

(1)

$$y' \sin x = y \ln y,$$

$$y = e \text{ when } x = \pi/3.$$

(2)

$$y dy + (xy^2 - 8x) dx = 0,$$

$$y = 3 \text{ when } x = 1.$$

Problem 9

Solve the following ODE

(1)

$$y'' - y' - 2y = 3e^{2x}$$

(2)

$$5y'' + 12y' + 20y = 120 \sin 2x$$

(3)

$$(D^2 + 2D + 17)y = 60e^{-4x} \sin 5x \quad \text{Hint: First solve } (D^2 + 2D + 17)y = 60e^{(-4+5i)x}.$$

(4)

$$5y'' + 6y' + 2y = x^2 + 6x$$

Problem 10

Solve the given differential equation by using the principle of superposition [see the solution of equation (6.29)].

$$(D - 1)^2 y = 4e^x + (1 - x)(e^{2x} - 1)$$